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A Comprehensive Report on Land Cover Classification of Abia-State Using Google Colab, Google Earth Engine, and Sentinel-2 Imagery with the Random Forest Algorithm

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1. Introduction

Land cover classification is a core remote sensing application used to identify and map surface features such as vegetation, built-up areas, water bodies, and bare land. Accurate land cover information supports urban planning, environmental monitoring, agricultural assessment, and resource management.

This project focused on producing a land cover classification map of Abia State using Sentinel-2 satellite imagery for the year 2024. The workflow combined Google Colab for notebook-based scripting and geospatial processing, Google Earth Engine for cloud-based satellite image analysis, and QGIS for shapefile preparation and validation.

The classification was carried out using the Random Forest supervised machine learning algorithm.

2. Aim of the Study

To generate a land cover classification map of Abia State using Sentinel-2 imagery and machine learning through an integrated workflow involving Google Colab, Google Earth Engine, GeoBoundaries, and QGIS.

3. Objectives

- To obtain the administrative boundary of Abia State from geoBoundaries.
- To process and convert the boundary data into shapefile format using QGIS.
- To upload the shapefile into Google Earth Engine as an Area of Interest (AOI).
- To acquire Sentinel-2 imagery for 2024.
- To derive spectral indices such as NDVI, NDBI, and NDWI.
- To generate training samples for land cover classes.

- To train a Random Forest classifier.
- To produce and export a final land cover classification map.
- To integrate notebook workflow execution using Google Colab.

4. Software / Tools Used

S/N	Tool	Purpose
1	Google Colab	Notebook environment for running geospatial scripts and project workflow
2	Google Earth Engine	Satellite imagery processing and classification

3	QGIS	Shapefile extraction, coordinate checks, spatial validation
4	geoBoundaries	Source of administrative boundary data
5	Sentinel-2 Imagery	Satellite data source

5. Detailed Methodology

5.1 Acquisition of Boundary Data

The administrative boundary of Abia State was downloaded from geoBoundaries, which provides open-access political and administrative boundaries globally.

The downloaded boundary data was obtained in vector format.

5.2 Boundary Preparation in QGIS

The downloaded Abia boundary dataset was opened in QGIS.

The following operations were carried out:

- Verified that the Abia State polygon was correctly identified.
- Checked geometry validity.
- Confirmed coordinate reference system.
- Extracted the required state boundary from larger administrative layers where necessary.
- Exported the selected boundary as an ESRI Shapefile.

The shapefile components included: .shp, shx, dbf, .prj

This prepared shapefile became the AOI dataset for further processing.

5.3 Upload of Shapefile into Google Earth Engine

The shapefile was uploaded into Google Earth Engine Assets as a project boundary.

The asset was referenced in script as:

```
var aoI = ee.FeatureCollection("projects/gee-abia-project/assets/Abia-State_Shp");
```

The study area was then visualized and centered on the map.

5.4 Google Colab Workflow Integration

Google Colab was used as the notebook environment to support coding workflow, script execution logic, and geospatial processing structure required by the lab exercise.

Colab formed part of the required academic workflow and was used to manage code cells, organize processing steps, and support reproducible analysis.

5.5 Acquisition of Sentinel-2 Imagery

Sentinel-2 Surface Reflectance imagery for the year 2024 was filtered over the Abia AOI using:

- Date range: 2024-01-01 to 2024-12-31
- Cloud cover threshold below 20%

To avoid incompatible image collection errors, only common spectral bands were selected:

- B2 (Blue)
- B3 (Green)
- B4 (Red)
- B8 (NIR)
- B11 (SWIR)

A median composite image was generated and clipped to the study area.

5.6 Derivation of Spectral Indices

Three indices were computed to improve class separation:

NDVI

Used to detect vegetation density.

NDBI

Used to identify built-up surfaces.

NDWI

Used to highlight water bodies.

These indices were stacked with Sentinel bands to create an 8-band predictor dataset.

5.7 Training Sample Creation

Manual training samples were created for four land cover classes using point features:

S/N	CLASS	CODE
1	Vegetation	0
2	Built-up	1
3	Water	2
4	Bareland	3

Twenty training points were generated across representative locations.

Where imported geometries existed as multi-points, they were converted into individual point features for model training.

5.8 Supervised Classification

A Random Forest classifier with 50 trees was trained using sampled pixel signatures extracted from the stacked image.

The classifier then predicted land cover classes across the entire study area.

5.9 Export of Final Output

The final classified raster map was exported from Google Earth Engine to Google Drive in GeoTIFF format for submission and future GIS use.

6. Major Challenges Encountered During Milestones

6.1 Boundary Preparation Challenge

Initial difficulty involved obtaining and preparing the correct Abia State boundary.

Solution: Boundary downloaded from geoBoundaries and refined/exported through QGIS.

6.2 Sentinel Image Compatibility Error

A heterogeneous image collection error occurred due to inconsistent bands.

Solution: Explicit selection of common Sentinel bands.

6.3 Training Sample Geometry Issues

Samples initially existed as multipoint geometry instead of individual features.

Solution: Converted to individual point features before sampling.

6.4 Variable Dependency Errors

Variables such as image, stacked, and trainingPoints were initially undefined due to ordering issues.

Solution: Rebuilt script in proper top-down sequence.

6.5 Export Delay

Final raster export required additional processing time.

Solution: Allowed cloud tasks to complete successfully.

7. Results

The final classified land cover map successfully distinguished:

- Vegetation
- Built-up areas
- Water bodies
- Bare land

The output was visually validated and exported successfully.

8. Conclusion

This project demonstrated an integrated modern geospatial workflow using Google Colab, Google Earth Engine, QGIS, and Sentinel-2 imagery to classify land cover in Abia State.

Despite technical hurdles encountered during preprocessing, training data structuring, and export stages, the project was completed successfully with a reliable final land cover product.



